

## ON THE KLEIN–GORDON EQUATION WITH THE SCREENED COULOMB POTENTIAL

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With the scalar and vector Coulomb + shifted Coulomb potentials  $V_{sc}(r) = \beta_{sc}/r + \gamma_{sc}/(r + \delta)$ ,  $\delta > 0$ , the Klein–Gordon equation is shown to possess at least one exact and elementary bound-state solution, provided that two of the couplings satisfy a pair of algebraic constraints. This complements the recent results obtained in the non-relativistic case.

In the non-relativistic Schrödinger equation the screened Coulomb interaction  $V(r) = \beta/r + \gamma/(r + \delta)$  proved to be useful, inter alia, for the description of heavy quarkonia, as one of the simplest “next-to-solvable” potentials [1]. In this context, it would be useful to introduce also (at least some of) the relativistic corrections in a non-perturbative way. Here we shall consider therefore the simple s-wave Klein–Gordon equation

$$\begin{aligned} & -(\mathrm{d}^2/\mathrm{d}r^2)\psi(r) + [m + V_1(r)]\psi(r) \\ & = [E - V_2(r)]^2\psi(r), \\ & V_i(r) = \beta_i/r + \gamma_i/(r + \delta), \quad i = 1, 2, \quad \delta > 0, \end{aligned} \quad (1)$$

with the modified coulombic form of both the scalar and vector potentials.

In direct analogy to the recent analysis of the non-relativistic case [2], we shall concentrate here on the problem of existence of the exact and simple solutions to (1).

*Theorem.* Eq. (1) possesses an elementary normalisable solution

$$\psi_e(r) = e^{-ar}(r + \delta)^cr^b, \quad a > 0, \quad b > \frac{1}{2}, \quad (2)$$

provided only that the couplings satisfy certain algebraic restrictions.

*Proof.* We may insert (2) into (1). Since

$$(\mathrm{d}/\mathrm{d}r)\psi_e(r) = [-a + c/(r + \delta) + b/r]\psi_e(r), \quad (3)$$

we obtain the condition

$$\begin{aligned} & -[-a + b/r + c/(r + \delta)]^2 + b/r^2 + c/(r + \delta)^2 \\ & + m^2 + 2m[\beta_1/r + \gamma_1/(r + \delta)] \\ & + [\beta_1/r + \gamma_1/(r + \delta)]^2 - E^2 \\ & + 2E[\beta_2/r + \gamma_2/(r + \delta)] \\ & - [\beta_2/r + \gamma_2/(r + \delta)]^2 = 0. \end{aligned} \quad (4)$$

This may be interpreted as a requirement that a fourth-order polynomial be equal to zero identically, i.e., eq. (1) will be satisfied by (2) if and only if the corresponding five coefficients (functions of the couplings  $\beta_1, \beta_2, \gamma_1, \gamma_2$  and  $\delta$ ) become equal to zero.

The first condition,

$$a^2 = m^2 - E^2, \quad (5)$$

is simple and corresponds to the  $r \rightarrow \infty$  limit of (4).

Similarly, we obtain the pair of requirements

$$\beta_1^2 - \beta_2^2 = b^2 - b, \quad \gamma_1^2 - \gamma_2^2 = c^2 - c, \quad (6)$$

when we consider  $r$  close to zero and  $-\delta$ , respectively.

Finally the remaining two requirements may be simplified by the re-parametrization of (5) and (6),

$$E = m \sin \lambda, \quad a = m \cos \lambda, \quad \lambda \in (0, \pi/2), \quad (7a)$$

$$\begin{aligned} \beta_1 &= \frac{1}{2} \operatorname{sh} \alpha, \quad \beta_2 = \frac{1}{2} \operatorname{ch} \alpha \cos \varphi, \\ b &= \frac{1}{2} + \frac{1}{2} \operatorname{ch} \alpha \sin \varphi, \quad \varphi \in (0, \pi), \end{aligned} \quad (7b)$$

$$\begin{aligned}\gamma_1 &= \frac{1}{2} \operatorname{sh} \rho, & \gamma_2 &= \frac{1}{2} \operatorname{ch} \rho \cos \chi, \\ c &= \frac{1}{2} + \frac{1}{2} \operatorname{ch} \rho \sin \chi, & \chi &\in (0, 2\pi),\end{aligned}\quad (7c)$$

and read

$$\begin{aligned}2 \cos \lambda + \operatorname{sh} \alpha + \operatorname{ch} \alpha \sin(\varphi - \lambda) \\ + \operatorname{sh} \rho + \operatorname{ch} \rho \sin(\chi - \lambda) = 0,\end{aligned}\quad (8)$$

and

$$\begin{aligned}1 + \operatorname{ch} \alpha \sin \varphi + \operatorname{ch} \rho \sin \chi - \operatorname{sh} \alpha \operatorname{sh} \rho \\ + \operatorname{ch} \alpha \operatorname{ch} \rho \cos(\varphi - \chi) \\ = (a\delta/\cos \lambda)[\operatorname{sh} \alpha - \operatorname{sh} \rho + \operatorname{ch} \alpha \sin(\varphi - \lambda) \\ - \operatorname{ch} \rho \sin(\chi - \lambda)].\end{aligned}\quad (9)$$

Real solutions to (8) and (9) exist — e.g., we may cancel  $V_1^2(r)$  and  $V_2^2(r)$  in (1) (by putting  $\beta_1 = \beta_2$  and  $\gamma_1 = \gamma_2$ ), obtain a Schrödinger-type equation and recall the results of refs. [1,2]. This completes the proof — in an appropriately specified domain, we may define  $\cos \lambda \in (-1, 1)$  by (8) and, after its insertion into (9), get the single algebraic requirement to be solved by the standard numerical techniques.

From a practical point of view, it seems reasonable

to re-interpret the parameters once more and to introduce  $\mu = \varphi - \lambda$  and  $\nu = \chi - \lambda$  instead of  $\varphi$  and  $\chi$ . Then, any four couplings  $\beta_{s,v}$  and  $\gamma_{s,v}$  are assigned the binding energy [pertaining to the elementary solution (2)], by (8). Similarly, eq. (9) becomes merely an explicit definition of the only unknown parameter  $\delta > 0$ . In this way, we avoid the need of computer completely.

We may conclude that at least some of the non-relativistic results pertaining to the potential  $V(r) = \beta/r + \gamma/(r + \delta)$  [1,2] may be extended to the relativistic case in a straightforward manner. This may prove to be useful not only in physical applications (inclusion of the relativistic corrections in model descriptions of the charged systems or, possibly, lighter quarkonia) but also in the purely methodical context.

### References

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